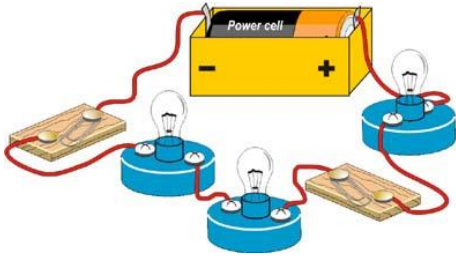




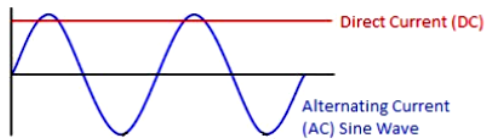
What is an Electrical Circuit?



Electrical circuit is a system of conductors through which current of electricity flows upon the application of electrical voltage. **DC Circuit** is an electrical circuit in which the applied source is DC. The common conductors in electrical circuits are **silver**, **copper** and **aluminum**.

Types of Electrical Current

There are two general types of electrical current, distinguished from each other by the manner in which they vary in magnitude and direction; these are classified as (1) **direct current** and (2) **alternating current**.



Direct Current

A **direct current**, sometimes called a **continuous current**, is an electric current that flows in one direction. One in which the energy transfer takes place unidirectionally, with changes in value from instant to instant that are either zero or so small that they may be neglected.

Direct current can be classified into three forms that can be derived from the three sources, namely: **continuous dc** which is produced by a battery, **unidirectional dc** derived from a dc generator and **pulsating dc** produced from rectifier circuits (either half-wave or full-wave).

Alternating Current

An **alternating current** is one in which the direction alternates regularly and, unless otherwise definitely stated, changes periodically in magnitude as well as direction. The two types of current flow are the **conventional flow** and the **electron flow**.

The **conventional flow of current** (+ to - direction) is still currently the preferred type of current flow. Most books are still using this flow. Unless otherwise stated or specified, the current is assumed to be conventional (from point of higher potential to point of lower potential).

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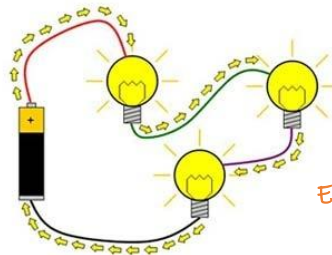
The **electron flow of current** which is opposite to the conventional flow is the actual flow of current. However, this flow is seldom considered since no matter what current flow is followed the magnitude of current remains the same.

Basic Electrical Circuit

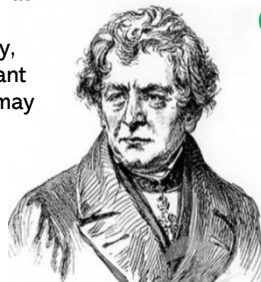
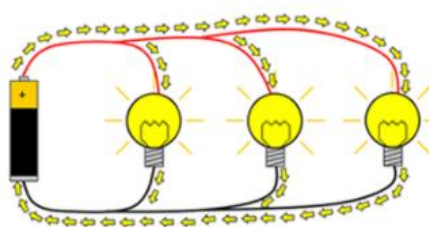
The basic circuit requirements are:

1. Source of potential difference (Battery, generator, power supply, etc).
2. Connecting wires (conductors) that provide the continuous path.

Series Circuit



Parallel Circuit



Ohm's Law

Ohm's law is the law that relates the three basic electrical quantities: current, voltage and resistance. This law is considered the fundamental law of an electrical circuit.

This was formulated in 1826 by **Georg Simon Ohm**. One simple statement of the law is:

"The current flowing in a circuit is directly proportional to the voltage (applied emf) and inversely proportional to the resistance."

The mathematical statement of the law is:

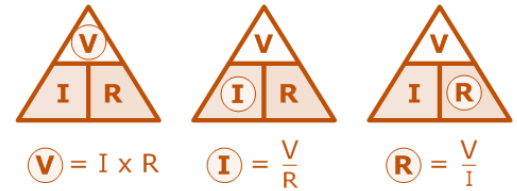
$$I \propto E$$

$$I = \frac{E}{R} = \frac{V}{R}$$

Note that Ohm's law is only applicable if the temperature is kept constant as temperature affects the resistance of a conductor. The resistance is always the constant ratio of voltage to current in Ohm's law.

The MAGIC TRIANGLE

The magic triangle (or the magic circle is some materials) is regarded as a good memory-aid for recalling Ohm's law.



Electrical Power

Generally, **power** is defined as the time rate at which work is done or the time rate of doing work hence electrical power is the time rate at which charged Q is forced to move by the applied voltage E.

$$P = \frac{W}{t} = \frac{Q}{t} E = EI$$

The unit of electrical power is **watt (W)** named after James Watt (1736-1819) where $1 \text{ W} = 1 \text{ J/s} = 1 \text{ V (A)}$. 1 horsepower (hp) = 746 W.



Joule's Law

The **Joule's law** is about power dissipation of a resistance element in a circuit formulated by **James Prescott Joule** (1818-1889), a British physicist.

The law states that:

"Electrical power is dissipated in a resistance whenever it carries an electrical current. The power dissipated is directly proportional to the square of electrical current and resistance."

The law is mathematically represented by:

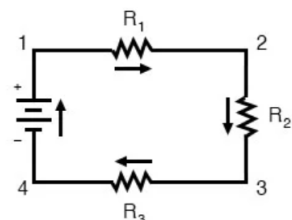
$$P \propto I^2$$

$$P = I^2 R = \frac{E^2}{R} = EI$$

Circuit Connections

Series Circuit

Series circuit is a circuit in which components like resistances are connected end to end so that there is only one path for current flow.



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Continue on Part 2